



Microbiology of the Oral Cavity

GIT-3rd year

LG

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Learning objectives

1

Describe the main ecological niches and resident microbiota of the oral cavity.

2

Explain how saliva, oxygen, pH, diet and tooth eruption shape oral microbial communities.

3

Explain dental plaque biofilm formation and its relationship to caries and periodontal disease.

4

Recognize the major microbial causes of acute oropharyngeal infection.

The oral cavity is a complex microbial ecosystem



Why the mouth is unique

It connects the external environment with the respiratory and gastrointestinal tracts.

The oral cavity contains the second-largest microbiota in the human body

Constant environmental change

Temperature, oxygen availability, pH, saliva flow and dietary exposures vary repeatedly throughout the day.

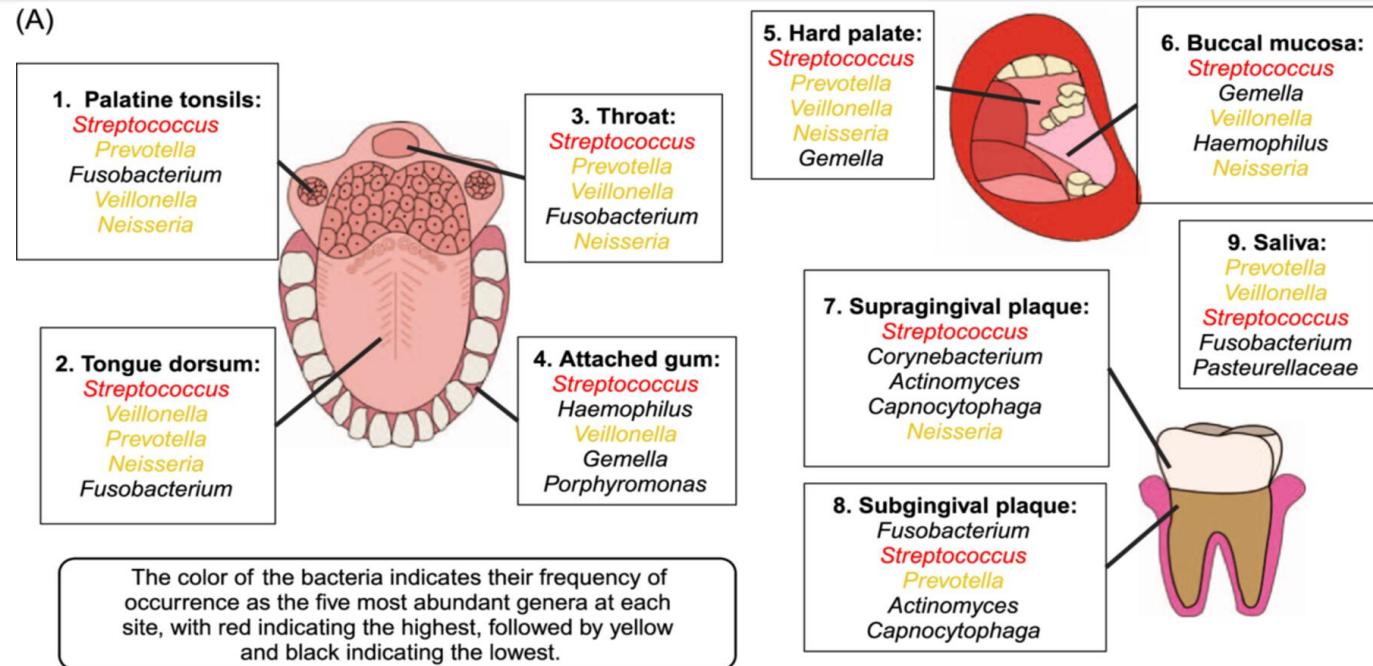
These changes create multiple microhabitats within a few centimeters.

Microbes live as communities

Oral cavity is complex and diverse ecosystem (consist of different microenvironment). Organisms interact, co-aggregate and compete rather than living as isolated species. It includes over 700 bacterial species, as well as fungi, protozoa, archaea, and viruses

Major oral microenvironments

- The oral cavity of a healthy individual is inhabited by multiple bacterial species.
- Most oral bacteria are either **anaerobic** or **facultatively anaerobic** and can thrive in the absence or presence of oxygen.
- Researchers classified the oral cavity into nine distinct areas based on their physical and chemical properties:



Classification of Oral Microflora

Resident microbiota

Communities of microorganisms that occur in oral cavity of all human.

Majority of these species are permanent flora

Usually lives in a commensal or mutualistic relationship with the host.

Helps provide colonization resistance and ecological stability.

Can become opportunistic when conditions change.

Transient microbiota

Acquired from food, water, hands and close contacts.

Often removed by resident competitors, immune defenses and saliva flow.

May occasionally establish infection if host barriers are compromised.

Core vs variable

Core: taxa commonly shared across healthy individuals.

Variable: The type of microorganisms in oral cavity that varies in different individuals and over time.

Both are shaped by age, diet, habits, medication and host biology.

Major members of the oral microbiota

Gram-positive bacteria

Viridans streptococci: *Streptococcus mutans*, *S. mitis*, *S. oralis*, *S. salivarius*, *S. sanguinis*

Actinomyces spp.

Peptostreptococcus and related anaerobic cocci

Gram-negative / anaerobic groups

Veillonella spp.

Prevotella

Porphyromonas

Fusobacterium spp.

Capnocytophaga spp.

Neisseria spp.

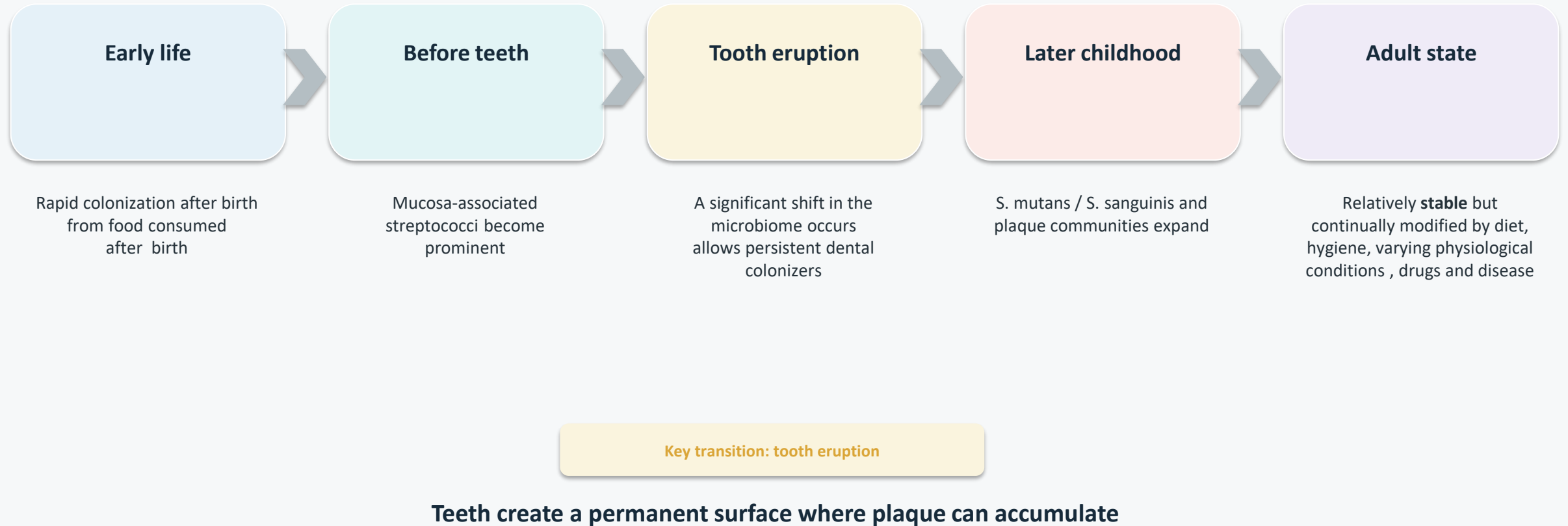
Other colonizers

Candida spp. may be carried without disease.

Viruses may infect, persist or reactivate in oral or tonsillar tissues.

Important point: presence ≠ disease.

Establishment of the oral microbiota across life



Saliva: transport medium and host defense



Physical effects

Continuous flow dilutes and removes microorganisms.
Saliva distributes microbes between oral surfaces.

Antimicrobial components

Secretory IgA
Lysozyme
Peroxidase system
Antimicrobial peptides

But surfaces still support heavy growth

Saliva itself is relatively poor in nutrients, yet food debris, epithelial material and gingival fluid create nutrient-rich local sites.
Therefore microbial growth is concentrated on surfaces such as teeth, tongue and gingival margins.

Factors that shape oral microbial communities

pH

acidification after
carbohydrate intake

Oxygen

highest at exposed surfaces;
lowest in gingival crevices

Diet

frequency and fermentable
carbohydrates matter

Saliva

Flow and antimicrobial
proteins

Oral hygiene

mechanical disruption of
plaque

Host immunity

mucosal and systemic
defenses

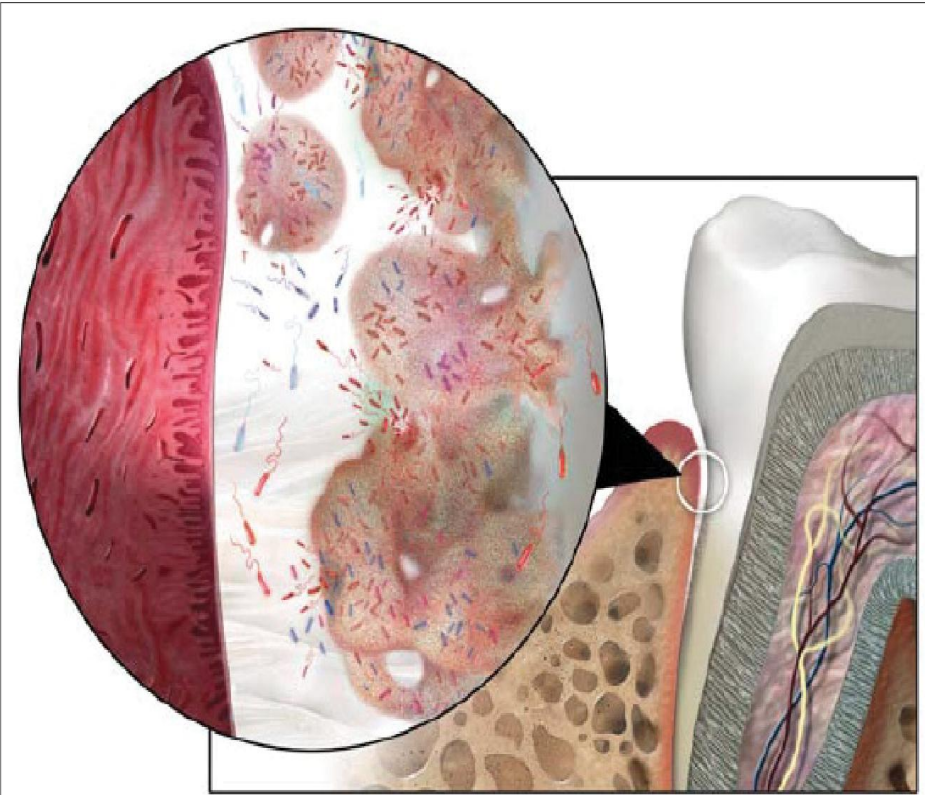
Drugs

antibiotics / steroids can
disturb balance

Age & habits

tooth status, smoking,
dentures, xerostomia

Dental plaque is a structured biofilm



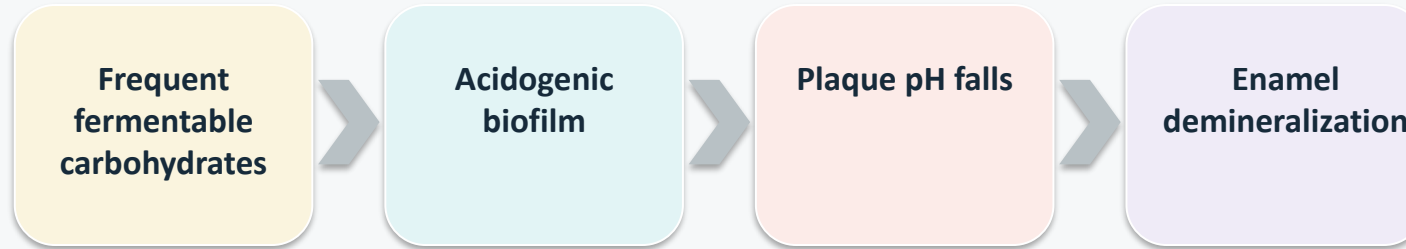
Pictured above: Plaque on teeth is a common example of biofilm.

- 1 Acquired pellicle
- 2 Early streptococcal adhesion
- 3 Co-aggregation
- 4 Maturation + matrix
- 5 Ecological shift / dysbiosis



Biofilm communities are embedded in an extracellular matrix and show cooperative interactions, persistence and altered susceptibility to host defenses.

Dental caries: an ecological acid-driven process

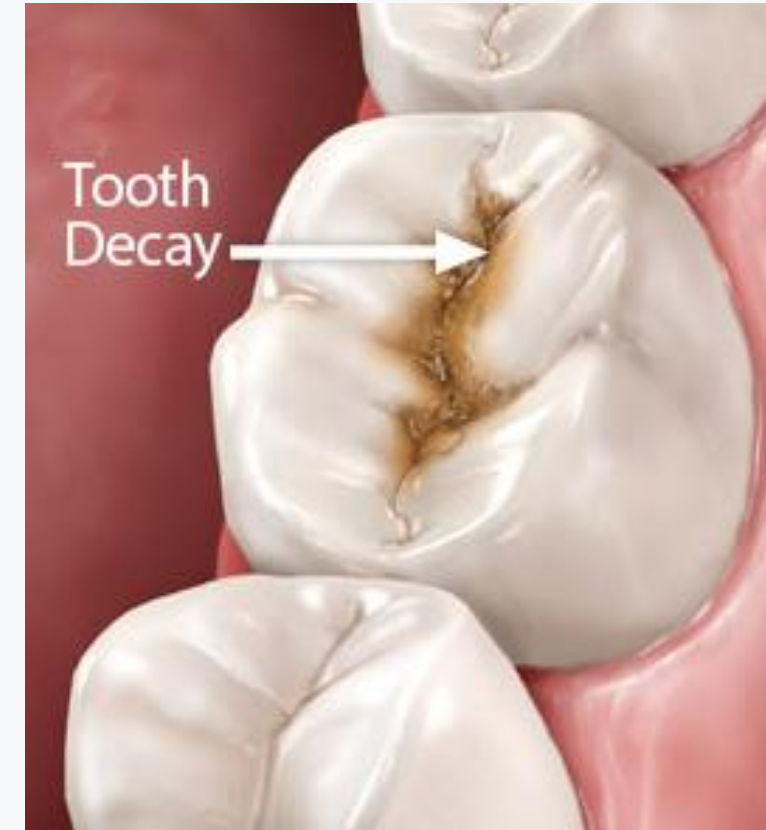


Microbial players

Streptococcus mutans group: major caries-associated oral streptococci and important plaque producers.

Lactobacillus spp. can contribute to acid production and progression in an established acidic environment.

Disease reflects the biofilm environment, not one organism acting alone.



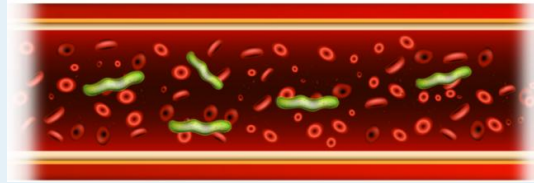
Why oral microbiology matters beyond the teeth

Local extension

Periodontal and dental infections may spread to jaw, sinus and facial spaces.

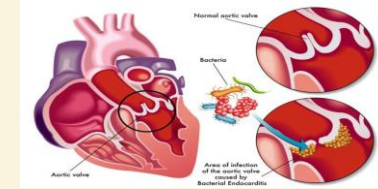
Bloodstream access

Mucosal trauma or dental procedures can cause transient bacteremia.



Endocarditis link

Viridans streptococci are classic oral organisms associated with subacute endocarditis.



Immunocompromised host

Oral Candida and invasive fungal lesions become more clinically important.

Oropharyngeal gateway

Tonsils and pharyngeal mucosa are sites of bacterial and viral infection and transmission.

Oropharyngeal infection: think syndrome first

SORE THROAT / PHARYNGITIS



Clinical appearance overlaps, so the goal is not to memorize a long organism list — it is to identify the few diagnoses that change management.

Bacterial causes

Streptococcus pyogenes (GAS)
S. dysgalactiae
Fusobacterium necrophorum
Corynebacterium diphtheriae
Neisseria gonorrhoeae

Viral causes

EBV
Adenovirus
Influenza / parainfluenza
Rhinovirus / coronavirus
Coxsackie A virus
HSV

Fungal / other

Candida spp. — especially with immune suppression or local risk factors
Ulcerative fungal disease in selected immunocompromised patients

Complications of GAS pharyngitis

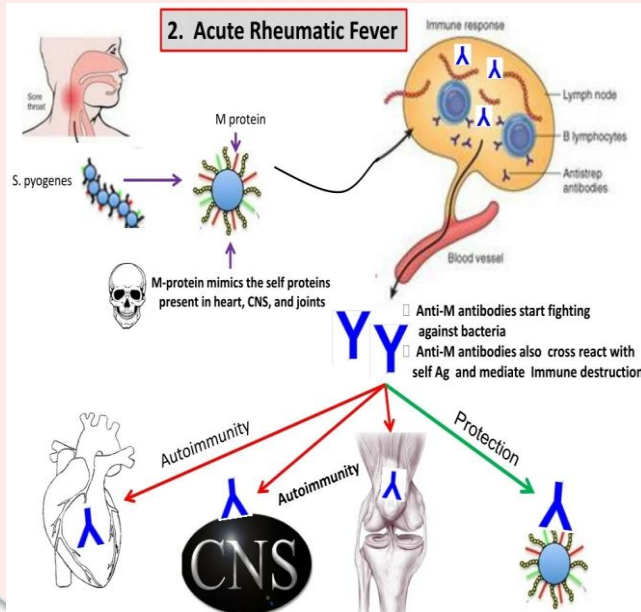
Suppurative complications

Peritonsillar infection / abscess
Cervical lymphadenitis
Otitis / sinus extension
Deep neck infection



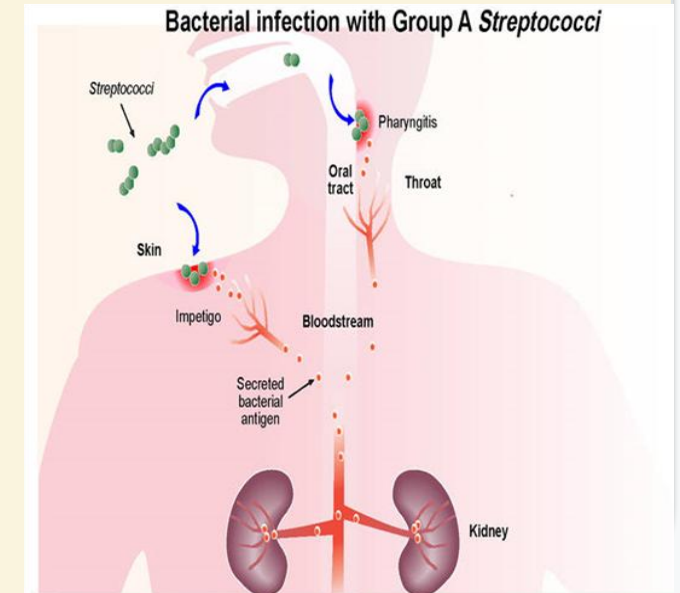
Acute rheumatic fever

Immune-mediated complication of streptococcal pharyngitis
Can involve heart, joints, skin and CNS
Timely treatment of GAS pharyngitis reduces risk



Acute glomerulonephritis

Immune-mediated glomerular inflammation
May follow pharyngeal or skin infection.
Edema, hypertension, hematuria and proteinuria



“Do not miss” bacterial alternatives to routine GAS pharyngitis

Diphtheria

Adherent gray pharyngeal pseudomembrane

Cervical swelling / “bull neck” may occur

Toxin-producing Corynebacterium diphtheriae causes systemic toxicity

Isolation, notification and urgent management are critical



Gonococcal pharyngitis

Consider with relevant sexual exposure

May be mild or asymptomatic

Requires site-appropriate microbiological testing and management



Oral candidiasis and viral oral lesions

Oral candidiasis (thrush)

Risk rises with immunosuppression, antibiotics, corticosteroids, diabetes, dentures and xerostomia.

Pseudomembranous thrush produces white plaques that can often be removed, leaving an erythematous surface.



Selected viral lesions

HSV: painful vesicles / ulcers; recurrent infection after latency.

Measles: Koplik spots on buccal mucosa.



Take-home messages

- 1 Oral microbiota are organized by place: teeth, mucosa, tongue, saliva, gingival crevice and oropharynx are ecologically different.
 - 2 Tooth eruption creates a non-shedding surface that fundamentally changes oral colonization and permits persistent plaque biofilm.
 - 3 Saliva, pH, oxygen, diet, hygiene and host defenses continually shape the oral microbial community.
 - 4 Dental caries and periodontal disease are biofilm-associated ecological diseases rather than simple single-organism infections.
 - 5 In pharyngitis, GAS is the key bacterial diagnosis .
 - 6 Remember important alternatives: diphtheria and oral candidiasis.
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References

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